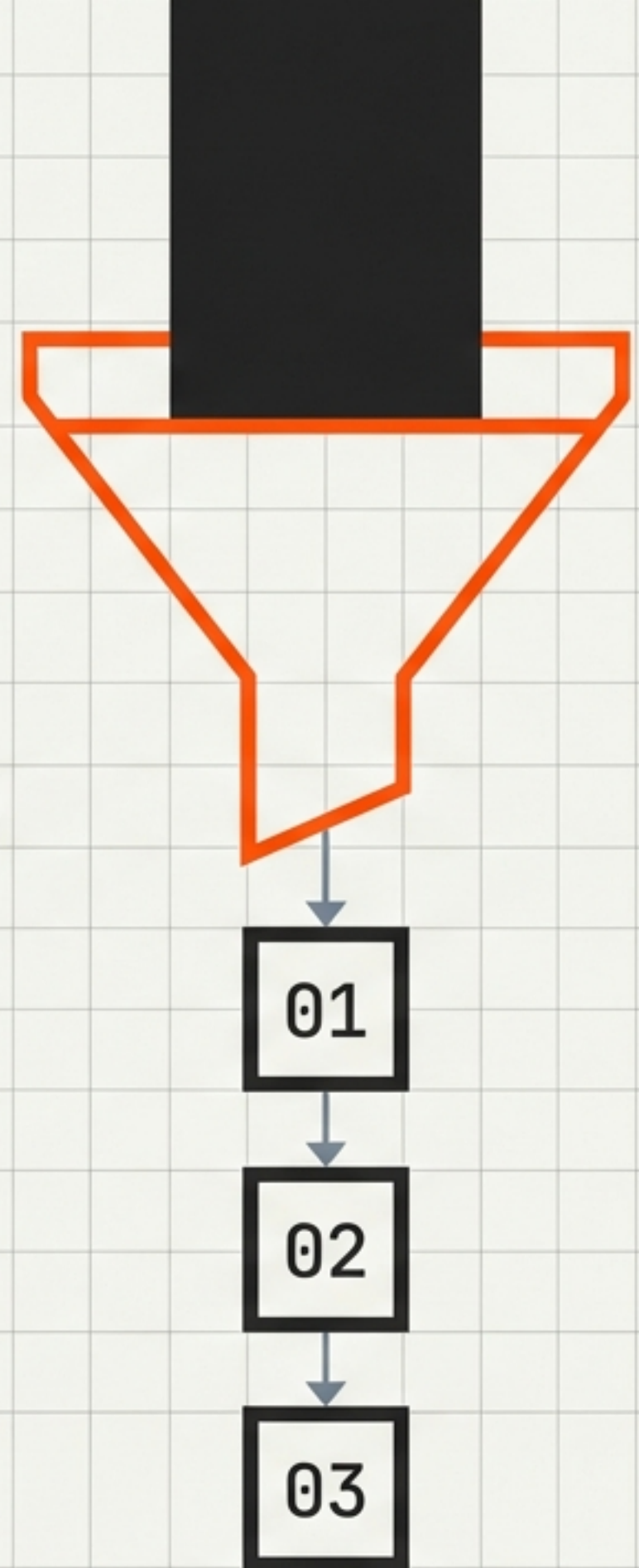


RLC Segmentation: Protocol Mechanics & Simulation

A technical deep dive into 3GPP TS 38.322 and ns-3 implementation strategies.

Context: Bridging the gap between theoretical HTML definitions and C++ test case validation.



Adapting Data for the Air Interface



The Concept

RLC (Radio Link Control) segmentation adapts large upper-layer packets to fit the constraints of the transmission medium.



The Mechanism

Large Service Data Units (SDUs) are sliced into Protocol Data Units (PDUs) using specific header fields (SI and SO) to ensure order.



The Verification

Logic is validated using the ns-3 network simulator and C++ test cases against 3GPP standards.

The Constraint: SDU vs PDU

SDU (Service Data Unit)

Original Payload /
Upper Layer

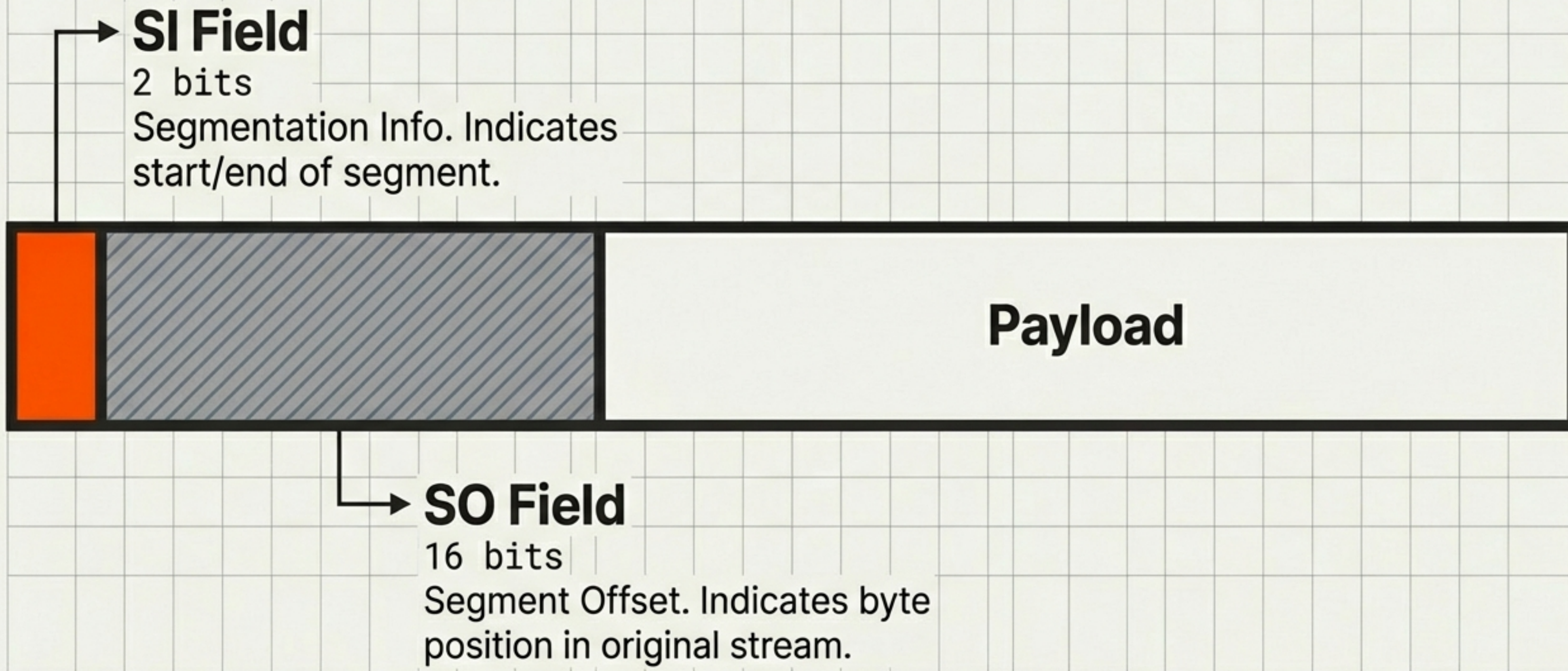
Size Mismatch

PDU (Protocol Data Unit)

Air Interface
Limit

Key Insight: The Maximum PDU size dictates the segmentation logic. The system must chop the SDU to fit.

Anatomy of the RLC Header



Decoding the Segment Offset (SO)



Segment 1 Header:
 $SO = 0$

Segment 2 Header:
 $SO = 500$

Purpose: Acts as a GPS for data.
Tells the receiver the exact byte position.

Logic:

1. **First Segment:** $SO = 0$.
2. **Subsequent:** Previous $SO + \text{Segment Size}$.

Field Length: 16 bits (allows large offsets).

Scenario: The 'Airport Pickup' SDU

Hey! Just wanted to let you know that I'll be arriving at the airport around 3 PM tomorrow. Can you pick me up? Thanks!



Transmission Slot

Transmission Slot

Transmission Slot

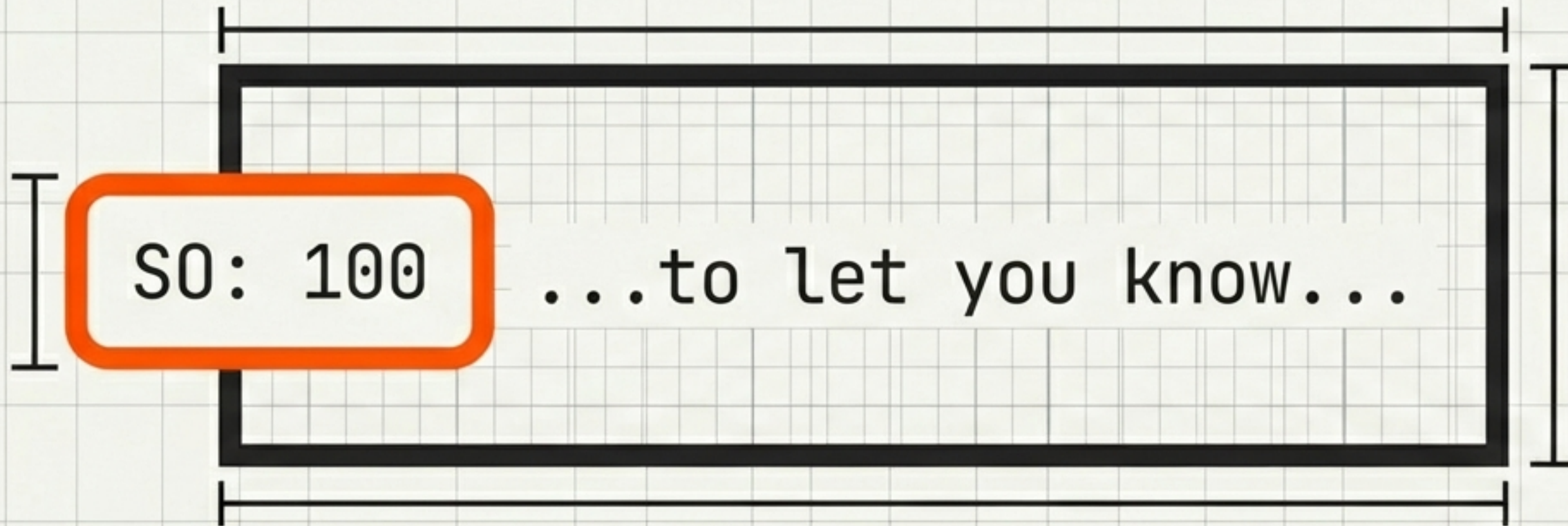
The Problem: The message SDU exceeds the available PDU size.

Segment 1: The Start of the Stream



- The SI field "01" indicates the Start.
- The SO field "0" tells the receiver to write this data at the very beginning of the buffer.

Segment 2: Calculating the Offset



- Byte Position: 100
- The receiver counts 100 bytes into the buffer and places this string there. Space is reserved for any missing bytes before this point.

From Theory to Verification: The ns-3 Framework



- **Objective:** Watch Real-Time Segmentation in Action.

Context: Simulating LTE/5G stacks to verify compliance with 3GPP specifications.

The Test Case Architecture

Access to Cellular
Protocol Stacks

```
#include "ns3/core-module.h"  
#include "ns3/lte-module.h"
```

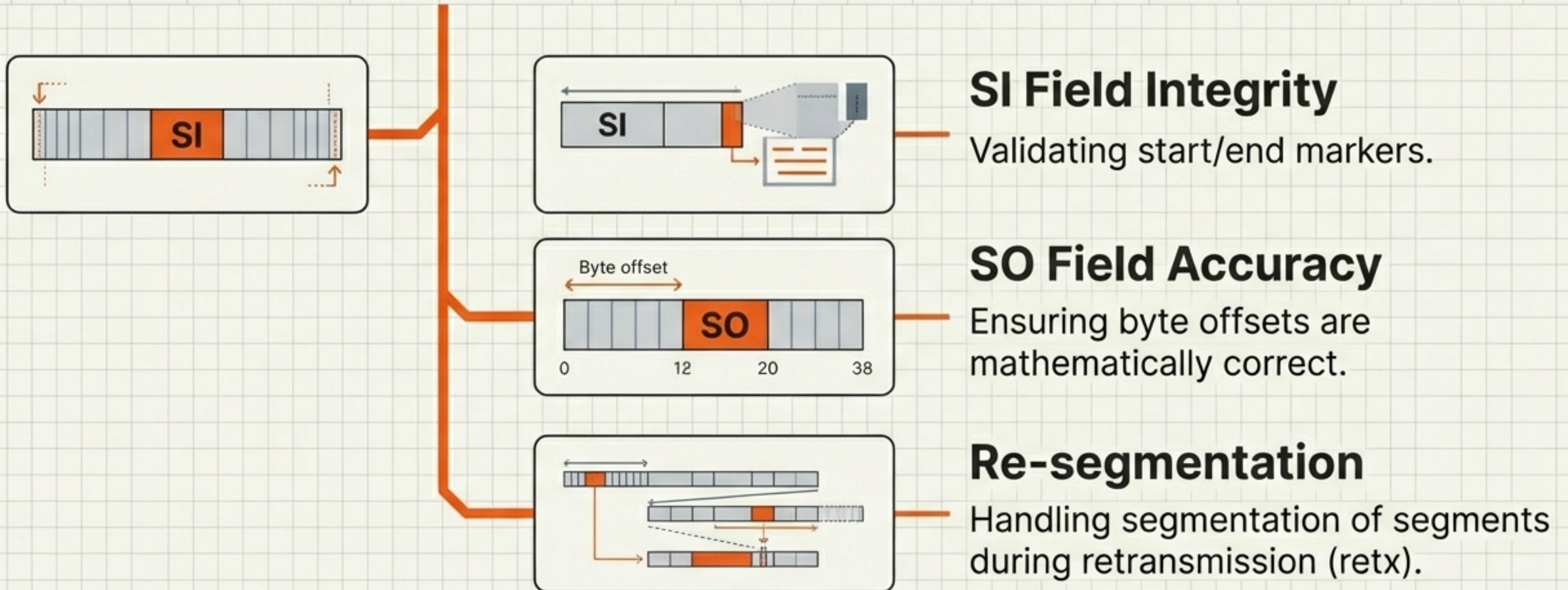
```
using namespace ns3;
```

```
class RlcSegmentationTest : public TestCase  
{  
    ...  
}
```

Integration with
ns-3 Test Suite
in Inter

Verification Objectives

Based on RLC Test Case 4 Code Logic



Compliance Target: 3GPP TS 38.322 Section 5.2

Edge Case: Re-segmentation on Retransmission

Segment X (Failed)



Re-segmentation
(Due to poor radio conditions)



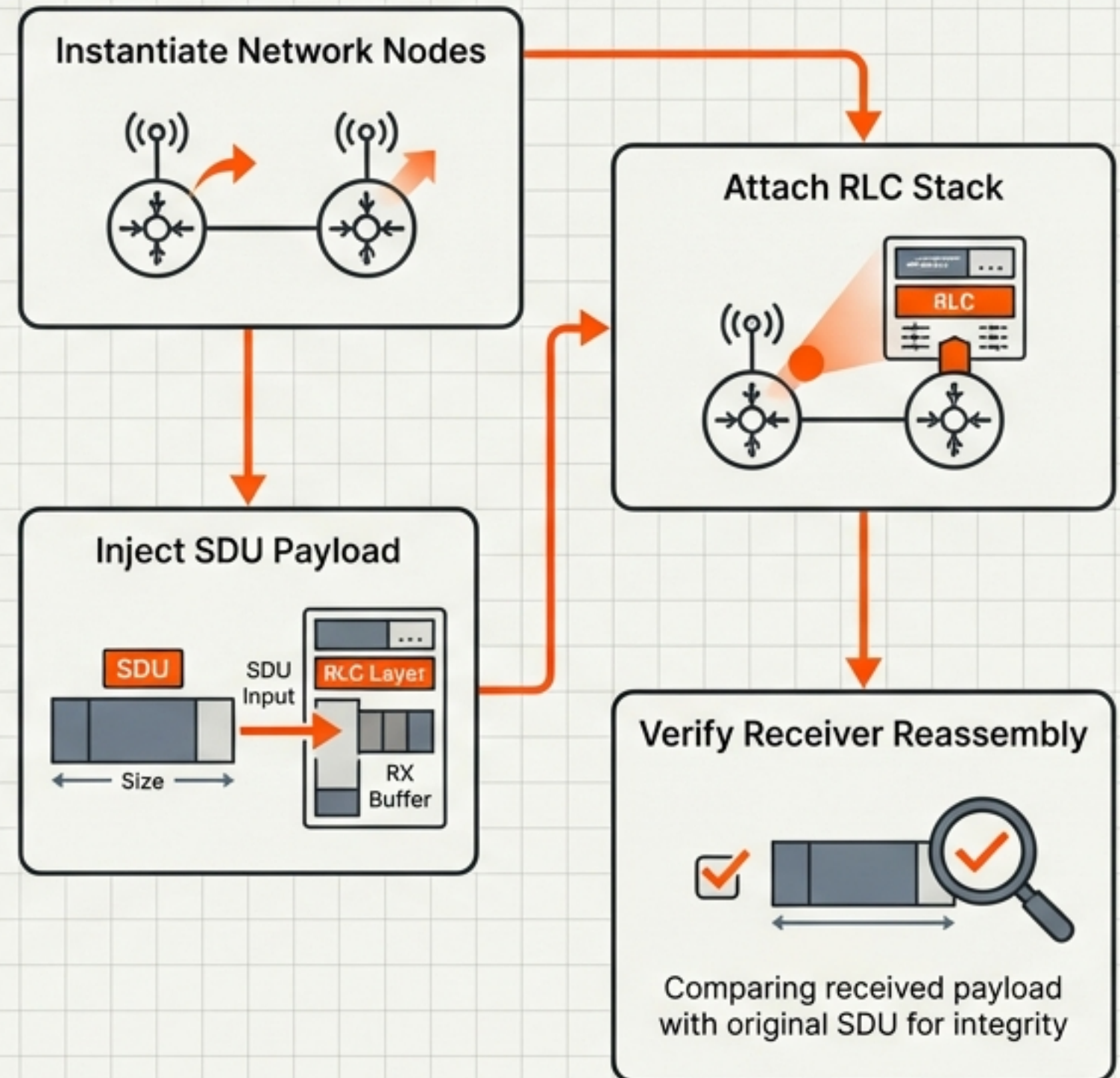
Scenario: A segment fails and must be re-sent, but the new radio slot is smaller.

Role of SO: The offset field must now track the position of a “fragment of a fragment”.

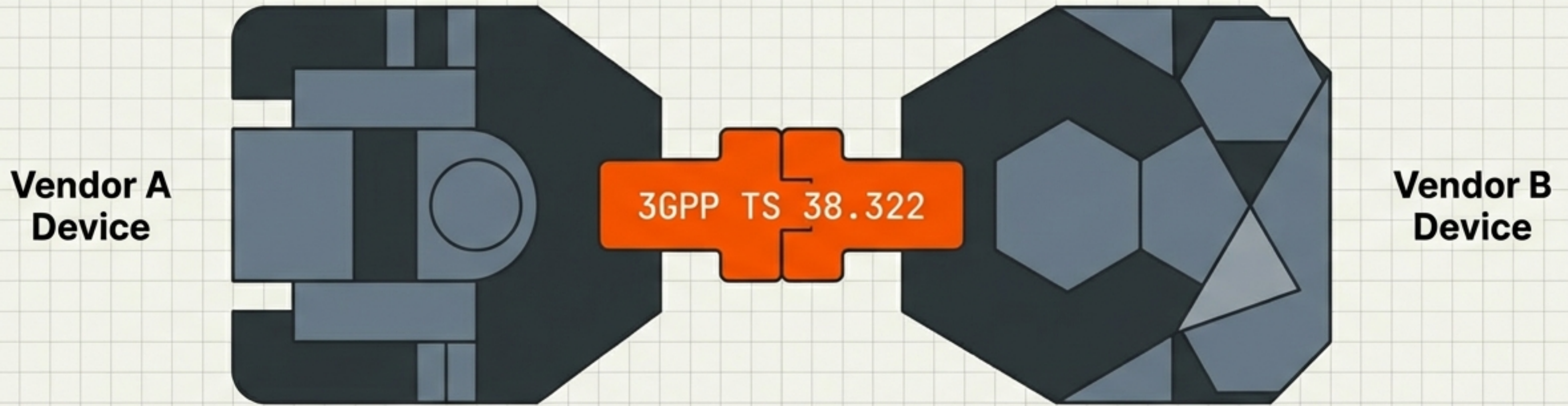
Executing the Simulation Logic

Based on RLC Test Case 4 Code Logic

```
public:  
  RlcSegmentationTest() : TestCase("RLC  
    Segmentation Test") {}  
  
  void DoRun() override  
  {  
    Ptr...
```



Why Standards Compliance Matters



The Trigger

SDUs are sliced based on real-time PDU limits.

The Map

16-bit SO fields provide the instructions for reassembly.

The Proof

ns-3 C++ test cases validate this logic against the standard.

References & Documentation

Standard Specification

3GPP TS 38.322 - Radio Link Control (RLC) Protocol Specification

Simulation Tool

ns-3 Network Simulator (LTE Module Documentation)

Source Code

segmentation_test_case.cc - RLC Test Case 4

Effective segmentation is the invisible engine that allows massive data streams to traverse the variable constraints of the cellular air interface.